

MATH 2551-D MIDTERM 3

VERSION A

FALL 2024

COVERS SECTIONS 15.5-8, 16.1-8

EXAM KEY

Honor code statement: I will abide strictly by the Georgia Tech honor code at all times. I will not use a calculator. I will not reference any website, application, or other CAS-enabled service. I will not consult with my notes or anyone during this exam. I will not provide aid to anyone else during this exam.

() **Initial here to attest to your integrity.**

Read all instructions carefully before beginning.

- Print your name and GT ID neatly above.
- You have 75 minutes to take the exam.
- You may not use aids of any kind.
- Show your work. Answers without work shown will receive little or no credit.
- The Learning Outcomes for the exam are listed on the back of this front cover.
- The Formula Sheet is on the final page and may be removed. Do not remove any other pages. Do not write any work on the Formula Sheet
- Good luck! Write yourself a message of encouragement on the front page!

Learning Outcomes

- **I3: Polar, Cylindrical, and Spherical Integrals.** Use polar, cylindrical, and spherical coordinates to transform double and triple integrals. Sketch regions based on given polar, cylindrical, and spherical iterated integrals.
- **I4: Triple Integrals.** Set up triple integrals as iterated integrals over any region. Sketch regions based on a given iterated integral. Interpret geometric meanings of triple integrals.
- **I5: Integral Applications.** Use multiple integrals to solve physical problems, such as finding area, average value, volume, mass of a lamina, or solid.
- **I6: Change of Variables.** Use general change of variables to transform double and triple integrals for easier calculation. Choose the most appropriate coordinate system to evaluate a specific integral.
- **V1: Scalar Line Integrals.** Define line integrals of a scalar function over a smooth curve. Evaluate scalar line integrals in two and three dimensions.
- **V2: Vector Line Integrals.** Define line integrals of a vector-valued function or vector field over a smooth curve. Evaluate vector line integrals in two and three dimensions.
- **V3: Conservative Vector Fields.** Test for conservative vector fields. Find potential functions. Evaluate line integrals using the Fundamental Theorem of Line Integrals.
- **V4: Green's Theorem.** Compute two-dimensional curls and divergence of vector fields. Evaluate integrals using Green's theorem. Solve applied problems using Green's Theorem, such as finding area.
- **V5: Surface Integrals.** Define and compute surface integrals for scalar and vector valued functions.
- **V6: Applications of Vector Calculus.** Interpret work, flow, flux, and surface area in terms of line and/or surface integrals, as appropriate.
- **V7: Generalizations of the FTC.** State and apply Stokes' Theorem and the Divergence Theorem to solve problems in three dimensions.

For problems 1-2 choose whether each statement is true or false. If the statement is *always* true, pick true. If the statement is *ever* false, pick false. Be sure to neatly fill in the bubble corresponding to your answer choice.

1. **[V6: Applications of Vector Calculus]** If $\mathbf{F} \cdot \mathbf{T} > 0$ at every point on a curve C , then the net flow of $\mathbf{F}$ along the curve C is positive.

☒ **TRUE**

☐ **FALSE**

2. **[V3: Conservative Vector Fields]** Let $\mathbf{F}$ be a conservative vector field in $\mathbb{R}^3$. If f and g both satisfy $\nabla f = \nabla g = \mathbf{F}$, then $f = g$.

☐ **TRUE**

☒ **FALSE**

3. **[V3, V4, V7]** Let $\mathbf{F}(x, y, z) = \langle 4x, -8y, 4z \rangle$ and S be the surface which is the part of the cylinder $x^2 + z^2 = 4$ between $y = 2024$ and $y = 2025$, oriented away from the y -axis. $\mathbf{F}$ is the curl of a vector field $\mathbf{F}$ since $\nabla \cdot \mathbf{F} = 0$. Select all of the integration theorems below that are appropriate to use to compute the flux of $\mathbf{F}$ across S .

☐ **A) Fundamental Theorem of Line Integrals**

☐ **B) Green's Theorem (circulation)**

☐ **C) Green's Theorem (flux)**

☒ **D) Stokes' Theorem**

☐ **E) Divergence Theorem**

4. **[V3, V4, V7]** Let $\mathbf{F}(x, y) = \langle 3x + y, x - 3y \rangle$ and C be the circle $x^2 + y^2 = 25$ oriented clockwise. Select all of the integration theorems below that are appropriate to use to compute the circulation of $\mathbf{F}$ along C .

☒ **A) Fundamental Theorem of Line Integrals**

☒ **B) Green's Theorem (circulation)**

☐ **C) Green's Theorem (flux)**

☐ **D) Stokes' Theorem**

☐ **E) Divergence Theorem**

5. [**I6: Change of Variables**] Suppose we wish to evaluate the integral $\iint_R f(x, y) \, dA$ using the change of variables

$$u = 3x \quad v = y^5.$$

Which of the following is the new area element dA in terms of du and dv ?

- ☐ A) $dA = 15v^{4/5} \, du \, dv$
- ☐ B) $dA = 15v^4 \, du \, dv$
- ☒ C) $dA = \frac{1}{15}v^{-4/5} \, du \, dv$
- ☐ D) $dA = \, du \, dv$
- ☐ E) $dA = \frac{5}{3}v^{-4/5} \, du \, dv$

6. [**V3: Conservative Vector Fields**] If possible, find a potential function for the vector field

$$\mathbf{F}(x, y) = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$$

with domain $x, y > 0$.

- ☒ A) $f(x, y) = \arctan(y/x)$
- ☐ B) $f(x, y) = \arctan(y/x) + \arctan(x/y)$
- ☐ C) $f(x, y) = -\frac{1}{2} \ln(x^2 + y^2)$
- ☐ D) $f(x, y) = \frac{1}{2} \ln(x^2 + y^2)$
- ☐ E) This vector field is not conservative.

7. [I3: Polar, Cylindrical, and Spherical Integrals] Which integral below computes the volume of the region inside the sphere $x^2 + y^2 + z^2 = 4$ and outside the cylinder $x^2 + y^2 = 2$?

- ☐ A) $\int_0^{2\pi} \int_{\sqrt{2}}^2 \int_0^{\sqrt{4-r^2}} dz \, dr \, d\theta$
- ☐ B) $\int_0^{\pi} \int_0^2 \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r \, dz \, dr \, d\theta$
- ☐ C) $\int_0^{2\pi} \int_0^{\pi/4} \int_{\sqrt{2} \csc(\phi)}^2 \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$
- ☐ D) $\int_0^{2\pi} \int_{\pi/4}^{3\pi/4} \int_0^2 \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$
- ☒ E) $\int_0^{2\pi} \int_{\pi/4}^{3\pi/4} \int_{\sqrt{2} \csc(\phi)}^2 \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$

8. [V5: Surface Integrals] Select all of the parameterizations $\mathbf{r}(s, t)$ below corresponding to the surface which is the part of the elliptical paraboloid $x = y^2 + 4z^2$ with $0 \leq x \leq 4$.

- ☐ A) $\mathbf{r}(s, t) = \langle s, t, s^2 + 4t^2 \rangle, \quad 0 \leq s^2 + 4t^2 \leq 4$
- ☒ B) $\mathbf{r}(s, t) = \langle s^2 + 4t^2, s, t \rangle, \quad 0 \leq s^2 + 4t^2 \leq 4$
- ☐ C) $\mathbf{r}(s, t) = \langle s \cos(t), s \sin(t), s^2 \cos^2(t) + 4s^2 \sin^2(t) \rangle, \quad 0 \leq s \leq 4, 0 \leq t \leq 2\pi$
- ☒ D) $\mathbf{r}(s, t) = \langle s, \sqrt{s} \cos(t), \frac{\sqrt{s}}{2} \sin(t) \rangle, \quad 0 \leq s \leq 4, 0 \leq t \leq 2\pi$
- ☒ E) $\mathbf{r}(s, t) = \langle 4t^2, 2t \cos(s), t \sin(s) \rangle, \quad 0 \leq s \leq 2\pi, 0 \leq t \leq 1$

9. [I4: Triple Integrals, I5: Integral Applications] Set up but do not compute an integral to compute the mass of the solid D that has mass density function $\delta(x, y, z) = 4 + \cos(x) \sin(y) + z/10$ and occupies the region

$$y \leq 8 - x^2, \quad x^2 + 2z^2 \leq y, \quad x \geq 0.$$

A complete solution will include a sketch of the shadow of D in the plane of the two variables that you do not integrate first.

Solution: The given constraints suggest integrating with respect to y first since we have

$$x^2 + 2z^2 \leq y \leq 8 - x^2.$$

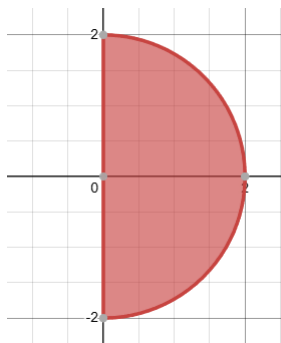
The shadow of this solid in the xz -plane is thus the region with

$$x^2 + 2z^2 \leq 8 - x^2 \text{ and } x \geq 0$$

i.e. the region

$$x^2 + z^2 \leq 4 \text{ and } x \geq 0.$$

A sketch of this region is given below; it is the right half of the disk of radius 2 centered at the origin in the xz -plane.



Therefore an integral to compute this mass is

$$\begin{aligned} \iiint_D \delta(x, y, z) \, dV &= \int_{-2}^2 \int_0^{\sqrt{4-z^2}} \int_{x^2+2z^2}^{8-x^2} 4 + \cos(x) \sin(y) + z/10 \, dy \, dx \, dz \\ \text{or } &\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{x^2+2z^2}^{8-x^2} 4 + \cos(x) \sin(y) + z/10 \, dy \, dz \, dx \\ \text{or } &\int_0^2 \int_{x^2}^{8-x^2} \int_{-\sqrt{(y-x^2)/2}}^{\sqrt{(y-x^2)/2}} 4 + \cos(x) \sin(y) + z/10 \, dz \, dy \, dx \end{aligned}$$

10. [V5: Surface Integrals, V6: Applications of Vector Calculus, V7: Generalizations of the FTC.] Compute the flux of the vector field

$$\mathbf{F}(x, y, z) = \langle xy^2 + \frac{1}{3}x^3, yx^2 + e^{x \cos(z)}, y^2z + \sin(\arctan(x)) \rangle$$

out of the closed surface S which is the portion of the paraboloid $z = x^2 + y^2$ from $z = 0$ to $z = 4$ together with the disk $x^2 + y^2 \leq 4$ in the plane $z = 4$. Simplify your final answer.

Hint: Cylindrical coordinates may be useful.

Solution: The surface is closed so we can apply the Divergence Theorem. Therefore the flux is

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma &= \iiint_D \nabla \cdot \mathbf{F} \, dV \\ &= \iiint_D (y^2 + x^2) + (x^2) + (y^2) \, dV \\ &= \int_0^{2\pi} \int_0^2 \int_{r^2}^4 2r^3 \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 8r^3 - 2r^5 \, dr \, d\theta \\ &= \int_0^{2\pi} 2r^4 - \frac{r^6}{3} \Big|_0^2 \, d\theta \\ &= \int_0^{2\pi} 32 - \frac{64}{3} \, d\theta \\ &= \int_0^{2\pi} \frac{32}{3} \, d\theta \\ &= \frac{64\pi}{3}. \end{aligned}$$

11. [V2: Vector Line Integrals, V3: Conservative Vector Fields, V4: Green's Theorem] Consider the vector field $\mathbf{F} = \langle 2y, x \rangle$ and the curve C which is the circle $x^2 + y^2 = 25$, oriented counterclockwise with outward normal vector.

(a) Compute $\int_C \mathbf{F} \cdot \mathbf{T} \, ds$.

Solution: We can apply the circulation form of Green's Theorem:

$$\begin{aligned} \int_C \mathbf{F} \cdot \mathbf{T} \, ds &= \iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} \, dA \\ &= \iint_R (1 - 2) \, dA \\ &= -\text{area}(R) \\ &= -25\pi. \end{aligned}$$

(b) Compute $\int_C \mathbf{F} \cdot \mathbf{n} \, ds$.

Solution: We can apply the flux form of Green's Theorem:

$$\begin{aligned} \int_C \mathbf{F} \cdot \mathbf{n} \, ds &= \iint_R (\nabla \cdot \mathbf{F}) \, dA \\ &= \iint_R (0 + 0) \, dA \\ &= 0 \end{aligned}$$

- (c) Based on your work above, what can you conclude about whether $\mathbf{F}$ is conservative? Justify your answer.

Solution: We know that $\mathbf{F}$ is not conservative because its circulation around the closed curve C is nonzero.

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FORMULA SHEET

Rules for Triple Integrals for the Sketching Impaired

- 1:** Choose a variable appearing exactly twice for the next integral.
- 2:** After setting up an integral, cross out any constraints involving the variable just used.
- 3:** Create a new constraint by setting the lower limit of the preceding integral less than the upper limit.
- 4:** A squared variable counts twice.
- 5:** The argument of a square root must be nonnegative.

- Trig identities: $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$, $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- Volume(D) = $\iiint_D dV$, $f_{avg} = \frac{\iiint_D f(x, y, z) dV}{\text{volume of } D}$, Mass: $M = \iiint_D \delta dV$
- Coordinate Transforms:
 - Cylindrical: $x = r \cos(\theta)$, $y = r \sin(\theta)$, $z = z$, $dV = r dz dr d\theta$
 - Spherical: $x = \rho \sin(\phi) \cos(\theta)$, $y = \rho \sin(\phi) \sin(\theta)$, $z = \rho \cos(\phi)$,
 $dV = \rho^2 \sin(\phi) d\rho d\phi d\theta$
- Substitution Theorem: If R is the image of G under a coordinate transformation $\mathbf{T}(u, v) = \langle x(u, v), y(u, v) \rangle$ then

$$\iint_R f(x, y) dx dy = \iint_G f(\mathbf{T}(u, v)) |\det D\mathbf{T}(u, v)| du dv.$$

- Line integrals:
 - $\int_C f(x, y, z) ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$
 - $\int_C \mathbf{F} \cdot \mathbf{T} ds = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$
 - $\int_C \mathbf{F} \cdot \mathbf{n} ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \langle y'(t), -x'(t) \rangle dt.$
- Fundamental Theorem of Line Integrals: $\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$ if C is a path from A to B
- Curl (Mixed Partial) Test: On a simply-connected region $\mathbf{F} = \nabla f \Leftrightarrow \text{curl } \mathbf{F} = 0$
- $\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$ $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$ $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$
- Green's Theorem: If C is a simple closed curve with positive orientation and R is the simply-connected region it encloses, then

$$\int_C \mathbf{F} \cdot \mathbf{T} ds = \iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} dA \qquad \int_C \mathbf{F} \cdot \mathbf{n} ds = \iint_R (\nabla \cdot \mathbf{F}) dA.$$

FORMULA SHEET CONTINUED

- Surface integrals:

$$- \iint_S f(x, y, z) \, d\sigma = \iint_R f(\mathbf{r}(u, v)) \, \|\mathbf{r}_u \times \mathbf{r}_v\| \, dA$$

$$- \iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iint_R \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA$$

- Stokes' Theorem: If S is a piecewise smooth oriented surface bounded by a piecewise smooth curve C and $\mathbf{F}$ is a vector field whose components have continuous partial derivatives on an open region containing S , then

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma.$$

- Divergence Theorem: If S is a piecewise smooth closed oriented surface enclosing a volume D and $\mathbf{F}$ is a vector field whose components have continuous partial derivatives on D , then

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \nabla \cdot \mathbf{F} \, dV.$$